

National Taiwan University
Department of Economics
ECON 4037: Game Theory
Fall 2026

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Office Hours: Wednesdays at 5:30 PM – 6:30 PM or by appointment

Class Time and Location

Wednesdays at 2:20 PM – 5:20 PM

Room 506, College of Social Sciences

Course Description

This course provides a rigorous introduction to game theory, the mathematical study of strategic interaction. It is designed primarily for advanced undergraduate students in their third or fourth years.

Game theory is a standard framework for analyzing strategic interaction across several disciplines, including economics, political science, biology, and law. The course aims to provide students with a foundational understanding of game-theoretic concepts that are applicable to real-world situations. By the end of this course, students will be able to:

- Formulate strategic situations as non-cooperative games in normal or extensive form;
- Analyze classic games such as the prisoner’s dilemma, models of oligopolistic competition, and job market signaling;
- Determine optimal strategies based on beliefs about others’ behavior;
- Apply core solution concepts to predict strategic behavior, including (i) iterated dominance, (ii) Nash equilibrium, (iii) mixed-strategy Nash equilibrium, (iv) subgame-perfect Nash equilibrium, (v) Bayesian Nash equilibrium, and (vi) perfect Bayesian equilibrium.

The prerequisites for this course are Microeconomics I and II (ECON 2018 and ECON 2019). Students are expected to be comfortable with calculus, probability, and statistics at the introductory level.

Course Requirements

Students are required to attend classes regularly, complete homework assignments, take the midterm exam, take the final exam, and submit a final essay. Attendance is expected but not tracked or directly graded (unless through exams). Students may discuss homework and the final essay with classmates or use generative AI tools while working on them. However, students must use classmates’ or AI assistance critically: they must understand, be able to defend, and take full responsibility for everything they write (for example, “AI wrote it” is not a valid excuse for errors or plagiarism). Exams are closed-book, meaning that AI and other outside assistance are not permitted.

Generative AI in Course Material Preparation

I use Claude Code (Anthropic’s Claude Sonnet 5 model) to assist me in preparing lecture slides,

homework problem sets, and exams, drawing on the Watson textbook and instructor resources. I direct this process closely, specifying the content, structure, and approach for each item. I review, revise, and finalize all such material myself, and I take full responsibility for any errors it contains.

Textbook

Watson, Joel. *Strategy: An Introduction to Game Theory*, 3rd ed., W. W. Norton & Company, 2013.

Grading

Homework (about 6 problem sets; lowest score dropped): 20%

Midterm Exam (Wednesday, October 28): 35%

Final Essay (due Wednesday, December 16): 10%

Final Exam (Wednesday, December 23): 35%

Final letter grades are assigned from the total course percentage according to NTU's official numeric grading scale. I may apply an upward curve.

A+	90–100	B+	77–79	C+	67–69
A	85–89	B	73–76	C	63–66
A-	80–84	B-	70–72	C-	60–62

A score of 60 (C-) is the minimum passing grade for the course. Scores from 50–59 (D) and below 50 (E) do not receive a passing grade.

Academic Integrity

Cheating on exams is strictly prohibited. Cheating includes, but is not limited to:

- Bringing unauthorized notes, study aids, or texts into the exam room, or consulting them outside the room during the exam (e.g., in the bathroom);
- Using a cell phone or other electronic device to store or access information during the exam;
- Having another person take the exam on your behalf;
- Communicating with classmates, looking at their work, or allowing others to look at your work during the exam.

In addition, students must avoid plagiarism in their essay. Plagiarism means presenting another person's work or ideas as your own without attribution; it does not include ideas generated from interacting with generative AI. Plagiarism includes, but is not limited to:

- representing another person's work or ideas as one's own;
- unacknowledged word-for-word use and/or paraphrasing of another person's work;
- inappropriate unacknowledged use of another person's ideas;
- submitting one's previous work (e.g. from a different class).

Absence Policy

Students who miss a homework deadline, the final essay deadline, or an exam due to illness, injury, a family emergency, or other serious circumstances beyond their control must report the

situation through myNTU as soon as possible, and in any case within three days of the missed date. Only absences reported this way will be considered for accommodation.

- **Homework:** Since the lowest homework score is already dropped, a single missed assignment needs no further accommodation. An excused absence from a second assignment extends that deadline by the length of the emergency, up to one week, unless other arrangements are made with the instructor in advance.
- **Final Essay:** An excused absence extends the deadline by the length of the emergency, up to one week, unless other arrangements are made with the instructor in advance.
- **Midterm Exam:** A student with an excused absence from the midterm will have their final exam score substituted for the midterm grade (the final then counts for 70% of the course grade). There is no separate makeup midterm.
- **Final Exam:** There is no makeup for the final exam except in extraordinary circumstances approved by the instructor in advance.

Course Schedule

The following is a tentative course schedule. Specific dates and topics are subject to adjustment.

Week	Date	Topic
Week 1	09/09	• Introduction (Ch. 1) • The Extensive Form (Ch. 2)
Week 2	09/16	• Strategies and the Normal Form (Ch. 3)
Week 3	09/23	• Beliefs, Mixed Strategies, and Expected Payoffs (Ch. 4) • General Assumptions and Methodology (Ch. 5)
Week 4	09/30	• Dominance and Best Response (Ch. 6) • Rationalizability and Iterated Dominance (Ch. 7)
Week 5	10/07	• Location, Partnership, and Social Unrest (Ch. 8) • Nash Equilibrium (Ch. 9)
Week 6	10/14	• Oligopoly, Tariffs, Crime, and Voting (Ch. 10)
Week 7	10/21	• Mixed-Strategy Nash Equilibrium (Ch. 11)
Week 8	10/28	Midterm Exam
Week 9	11/04	• Details of the Extensive Form (Ch. 14) • Sequential Rationality and Subgame Perfection (Ch. 15)
Week 10	11/11	• Repeated Games and Reputation (Ch. 22)
Week 11	11/18	• Random Events and Incomplete Information (Ch. 24)
Week 12	11/25	• Bayesian Nash Equilibrium and Rationalizability (Ch. 26)
Week 13	12/02	Instructor Travel – No class
Week 14	12/09	• Lemons, Auctions, and Information Aggregation (Ch. 27)
Week 15	12/16	• Perfect Bayesian Equilibrium (Ch. 28) Final Essay Due

Week	Date	Topic
Week 16	12/23	Final Exam